Every metric, wins and losses

Raw scores. Adopted method (attention, t = 9) and CLR, then three exploratory a3r variants (dashed), then competitors.

	Attention *adopted*	CLR	a3r *exploratory*	a3r2 *exploratory*	a3r3 *exploratory*	GENIE3	PIDC	GRNBoost2	|r|
E. coli, M3D compendium (466 experiments, RegulonDB)									
Regulons, strong evidence	1.68	1.44	1.72	● **1.73**	1.66	1.51	1.45	1.36	1.23
Regulons, all evidence	1.44	1.37	● **1.56**	1.51	1.46	1.40	1.40	1.26	1.23
Direct TF-to-target edges (AUPR ratio)	5.17	● **6.75**	5.84	3.95	3.30	6.54	5.53	5.17	3.74
Direct edges: median per-TF AUROC (chance 0.50)	0.622	0.595	0.627	0.581	0.568	● **0.654**	0.605	0.640	0.596
Per-regulon mean AP, strong evidence	● **25.6**	23.0	25.2	25.3	25.3	24.3	18.8	20.9	19.3
Per-regulon mean AP, all evidence	● **16.7**	14.8	16.6	16.4	15.9	16.4	12.3	14.1	12.2
Protein complexes (Complex Portal)	22.6	21.8	26.1	31.0	● **32.5**	–	–	–	29.2
E. coli, EcoMAC compendium (2,198 arrays)									
Regulons, strong evidence	1.68	1.39	1.72	● **1.77**	1.76	1.50	1.43	1.36	1.18
Regulons, all evidence	1.47	1.33	● **1.58**	1.57	1.47	1.38	1.39	1.26	1.19
Protein complexes (Complex Portal)	22.6	23.0	● **26.3**	–	–	–	–	–	26.0
Human K562 Perturb-seq (essential genes)									
DoRothEA A-B (TF targets)	1.09	1.06	● **1.11**	1.10	1.07	1.08	1.07	1.05	1.05
DoRothEA A-B-C (TF targets)	1.07	1.03	● **1.07**	1.06	1.05	1.04	1.04	1.03	1.03
CollecTRI (TF targets)	1.10	1.03	● **1.13**	1.09	1.05	1.04	1.05	1.04	1.01
CORUM protein complexes	3.51	6.39	7.95	● **8.19**	8.16	6.32	6.33	3.66	7.00
Knockdown-response modules	2.38	2.42	● **3.13**	2.57	2.13	2.45	2.35	1.84	2.45
Human RPE1 Perturb-seq									
CORUM protein complexes (merged)	3.33	● **6.77**	6.14	–	–	–	–	–	2.84
Single cells (BEELINE, 14 cases)									
Median AUPRC ratio, primary truths	1.54	1.48	● **1.73**	1.65	1.65	1.69	1.56	1.47	1.20
Median AUPRC ratio, cell-type ChIP truth	0.95	1.00	0.96	0.97	1.00	● **1.02**	0.98	1.02	1.02

Bars start at random (tick; 1.00, or 0.50 for the AUROC row) and are scaled within each row; the value, not the bar, carries the comparison across rows. Bold with a dot: highest value in the row.
Ties are not marked here (see the method table, docs/whitepapers/sota_table.md, for Bonferroni-tied cells). No intervals, no tests. A grey dash is a cell that was not run.
Pooled AUPR / base rate (1.00 = random), raw scores unless marked hub-controlled; BEELINE median over 14 cases; a3r was picked among about 20 read-outs by its K562 scores (selected, not fitted);
K562 and M3D are tuning sets; attention and a3r settings were developed on M3D; DREAM5 yeast is separate (page 2 / section 8) and was not run for a3r.
BEELINE: scores of CLR-NS for the attention and a3r columns (attention at its held-out depth, a3r family at t = 9, not tuned); CLR is plain CLR; median over 14 cases x BEELINE primary truths (44 case-truth cells), second row cell-type ChIP only.